



Chords and cyclic quadrilaterals

A CHORD IS A STRAIGHT LINE JOINING ANY TWO POINTS ON THE CIRCUMFERENCE OF A CIRCLE. A CYCLIC QUADRILATERAL HAS FOUR CHORDS AS ITS SIDES.

Chords vary in length—the diameter of a circle is also its longest chord. Chords of the same length are always equal distances from the center of the circle. The corners of a cyclic quadrilateral (four-sided shape) touch the circumference of a circle.

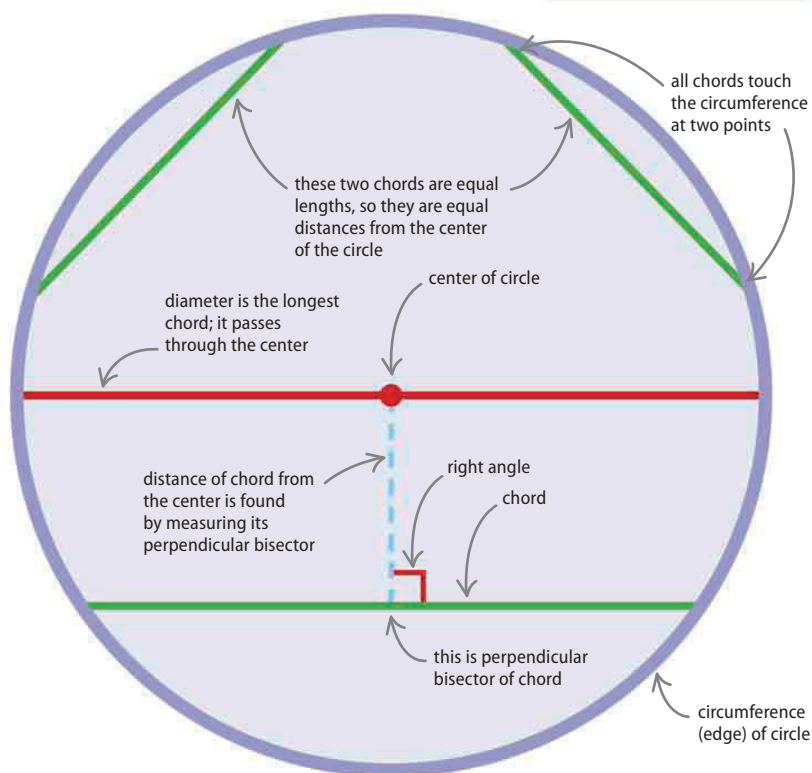
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Chords

A chord is a straight line across a circle. The longest chord of any circle is its diameter because the diameter crosses a circle at its widest point. The perpendicular bisector of a chord is a line that passes through its center at right angles (90°) to it. The perpendicular bisector of any chord passes through the center of the circle. The distance of a chord to the center of a circle is found by measuring its perpendicular bisector. If two chords are equal lengths they will always be the same distance from the center of the circle.



▷ Chord properties

This circle shows four chords. Two of these chords are equal in length. The longest chord is the diameter, and one is shown on the right with its perpendicular bisector (a line that cuts it in half at right angles).

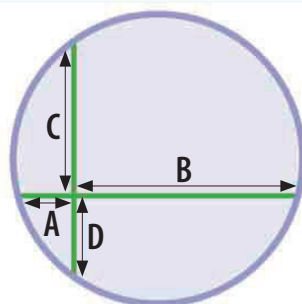
LOOKING CLOSER

Intersecting chords

When two chords cross, or “intersect,” they gain an interesting property: the two parts of one chord, either side of where it is split, multiply to give the same value as the answer found by multiplying the two parts of the other chord.

▷ Crossing chords

This circle shows two chords, which cross one another (intersect). One chord is split into parts A and B, the other into parts C and D.



two parts of one chord multiplied by each other

two parts of other chord multiplied by each other

$$A \times B = C \times D$$